

Oriented Surfaces

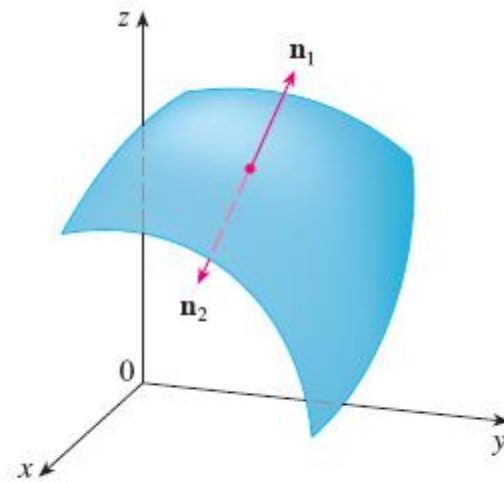


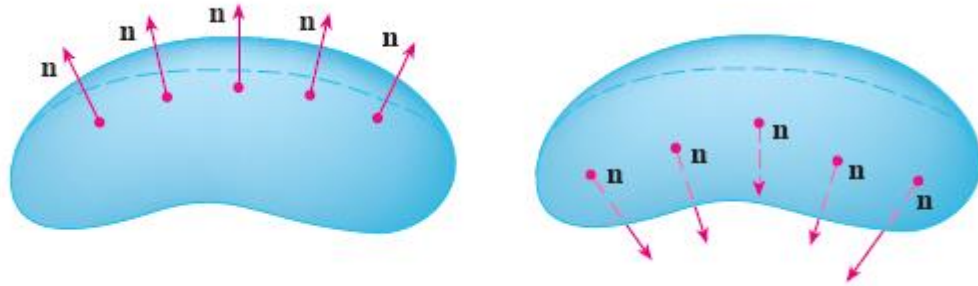
FIGURE 6

From now on we consider only orientable (two-sided) surfaces. We start with a surface S that has a tangent plane at every point (x, y, z) on S (except at any boundary point). There are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2 = -\mathbf{n}_1$ at (x, y, z) . (See Figure 6.)

If it is possible to choose a unit normal vector $\mathbf{n}$ at every such point (x, y, z) so that $\mathbf{n}$ varies continuously over S , then S is called an **oriented surface** and the given choice of $\mathbf{n}$ provides S with an **orientation**. There are two possible orientations for any orientable surface (see Figure 7).

FIGURE 7

The two orientations
of an orientable surface



Non Oriented Surface

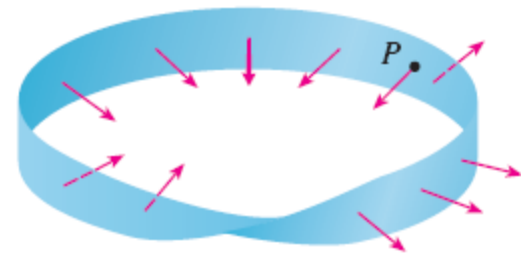
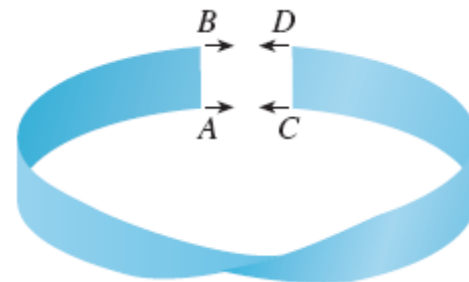


FIGURE 4

A Möbius strip



For a surface $z = g(x, y)$ given as the graph of g , we use Equation 3 to associate with the surface a natural orientation given by the unit normal vector

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$$\mathbf{n} = \frac{-\frac{\partial g}{\partial x} \mathbf{i} - \frac{\partial g}{\partial y} \mathbf{j} + \mathbf{k}}{\sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}}$$

Since the $\mathbf{k}$ -component is positive, this gives the *upward* orientation of the surface.

If S is a smooth orientable surface given in parametric form by a vector function $\mathbf{r}(u, v)$, then it is automatically supplied with the orientation of the unit normal vector

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$$\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|}$$

and the opposite orientation is given by $-\mathbf{n}$.

An interesting Example

the sphere $x^2 + y^2 + z^2 = a^2$.

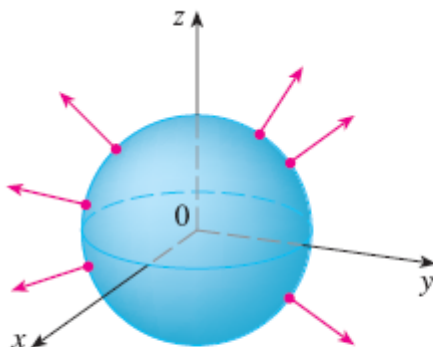


FIGURE 8
Positive orientation

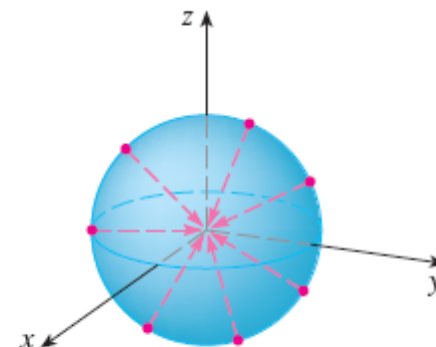
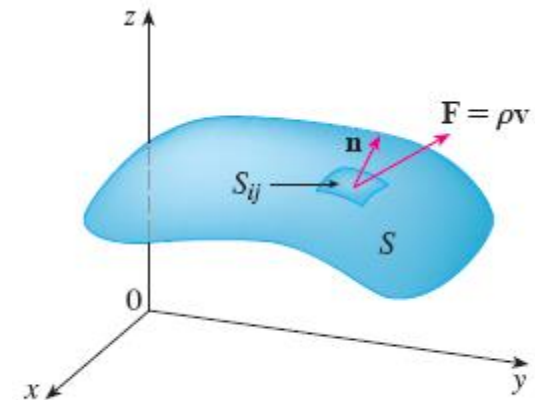


FIGURE 9
Negative orientation

For a **closed surface**, that is, a surface that is the boundary of a solid region E , the convention is that the **positive orientation** is the one for which the normal vectors point *outward* from E , and inward-pointing normals give the negative orientation (see Figures 8 and 9).

Surface Integrals of Vector Fields



Suppose that S is an oriented surface with unit normal vector $\mathbf{n}$, and imagine a fluid with density $\rho(x, y, z)$ and velocity field $\mathbf{v}(x, y, z)$ flowing through S . (Think of S as an imaginary surface that doesn't impede the fluid flow, like a fishing net across a stream.) Then the rate of flow (mass per unit time) per unit area is $\rho\mathbf{v}$. If we divide S into small patches S_{ij} , as in Figure 10 (compare with Figure 1), then S_{ij} is nearly planar and so we can approximate the mass of fluid per unit time crossing S_{ij} in the direction of the normal $\mathbf{n}$ by the quantity

$$(\rho\mathbf{v} \cdot \mathbf{n})A(S_{ij})$$

where ρ , $\mathbf{v}$, and $\mathbf{n}$ are evaluated at some point on S_{ij} . (Recall that the component of the vector $\rho\mathbf{v}$ in the direction of the unit vector $\mathbf{n}$ is $\rho\mathbf{v} \cdot \mathbf{n}$.) By summing these quantities and taking the limit we get, according to Definition 1, the surface integral of the function $\rho\mathbf{v} \cdot \mathbf{n}$ over S :

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$$\iint_S \rho \mathbf{v} \cdot \mathbf{n} \, dS = \iint_S \rho(x, y, z) \mathbf{v}(x, y, z) \cdot \mathbf{n}(x, y, z) \, dS$$

and this is interpreted physically as the rate of flow through S .

A more precise look at the rate of flow

Volume of fluid crossing dS in Δt seconds

= volume contained in cylinder of base dS and slant height $\mathbf{v} \Delta t$

$$= (\mathbf{v} \Delta t) \cdot \mathbf{n} \, dS = \mathbf{v} \cdot \mathbf{n} \, dS \, \Delta t$$

Then, volume per second of fluid crossing $dS = \mathbf{v} \cdot \mathbf{n} \, dS$

so that mass of fluid crossing dS per second = $(\rho \bar{v}) \cdot \hat{n} \, dS$,
 where $\rho(x, y, z)$ is the
 density of the
 fluid at the
 pt. (x, y, z) .

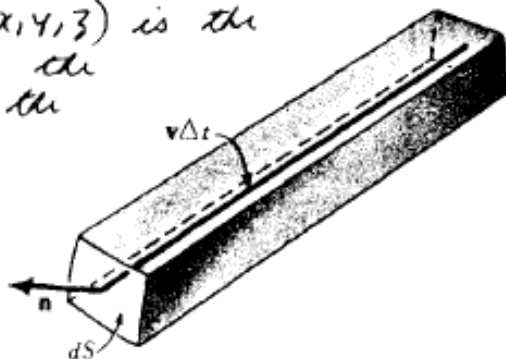


Fig. (a)

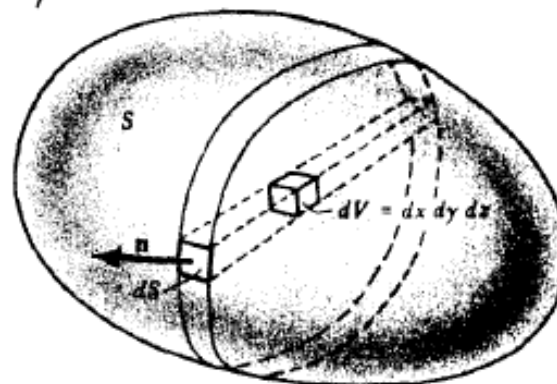


Fig. (b)



If we write $\mathbf{F} = \rho \mathbf{v}$, then $\mathbf{F}$ is also a vector field on $\mathbb{R}^3$ and the integral in Equation 7 becomes

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

A surface integral of this form occurs frequently in physics, even when $\mathbf{F}$ is not $\rho \mathbf{v}$, and is called the *surface integral* (or *flux integral*) of $\mathbf{F}$ over S .

8 Definition If $\mathbf{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\mathbf{n}$, then the **surface integral of $\mathbf{F}$ over S** is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

This integral is also called the **flux** of $\mathbf{F}$ across S .

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$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA$$

EXAMPLE 4 Find the flux of the vector field $\mathbf{F}(x, y, z) = z \mathbf{i} + y \mathbf{j} + x \mathbf{k}$ across the unit sphere $x^2 + y^2 + z^2 = 1$.

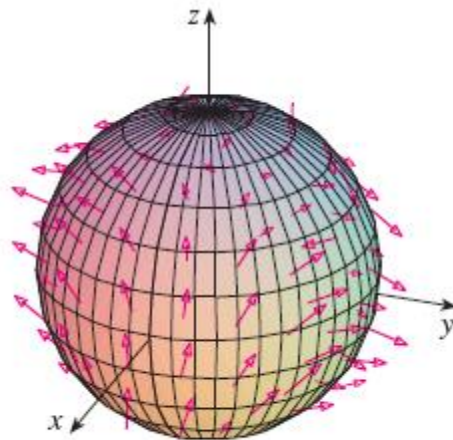


FIGURE 11

$$\frac{4\pi}{3}$$

Graphs

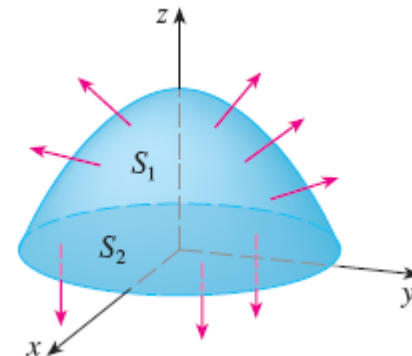
In the case of a surface S given by a graph $z = g(x, y)$,

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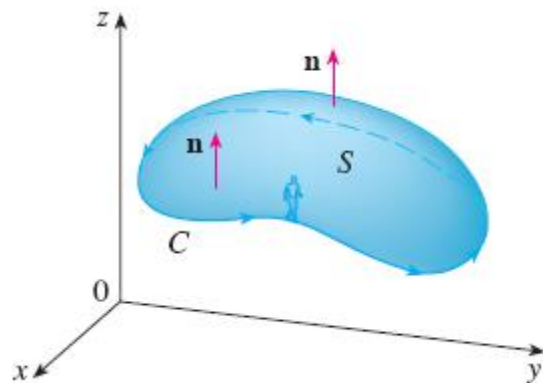
$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA$$

V EXAMPLE 5 Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + z\mathbf{k}$ and S is the boundary of the solid region E enclosed by the paraboloid $z = 1 - x^2 - y^2$ and the plane $z = 0$.

$$\frac{\pi}{2}$$



Stokes' Theorem



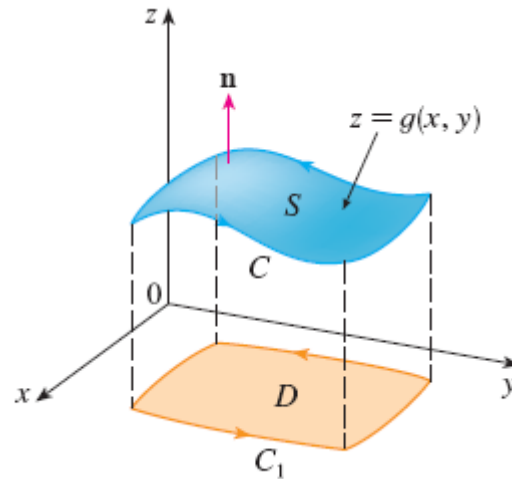
Stokes' Theorem Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

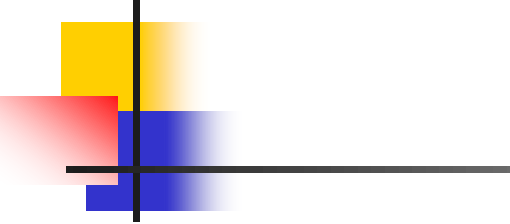
$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$

Stokes' Theorem says that the line integral around the boundary curve of S of the tangential component of $\mathbf{F}$ is equal to the surface integral over S of the normal component of the curl of $\mathbf{F}$.

There is an analogy among Stokes' Theorem, Green's Theorem, and the Fundamental Theorem of Calculus. As before, there is an integral involving derivatives on the left side of Equation 1 (recall that $\text{curl } \mathbf{F}$ is a sort of derivative of $\mathbf{F}$) and the right side involves the values of $\mathbf{F}$ only on the *boundary* of S .

PROOF OF A SPECIAL CASE OF STOKES' THEOREM We assume that the equation of S is $z = g(x, y)$, $(x, y) \in D$, where g has continuous second-order partial derivatives and D is a simple plane region whose boundary curve C_1 corresponds to C . If the orientation of S is upward, then the positive orientation of C corresponds to the positive orientation of C_1 . (See Figure 2.) We are also given that $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, where the partial derivatives of P , Q , and R are continuous.





$$\boxed{2} \quad \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$

$$= \iint_D \left[-\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \frac{\partial z}{\partial x} - \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \frac{\partial z}{\partial y} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \right] dA$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_a^b \left(P \frac{dx}{dt} + Q \frac{dy}{dt} + R \frac{dz}{dt} \right) dt \\ &= \int_a^b \left[P \frac{dx}{dt} + Q \frac{dy}{dt} + R \left(\frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \right) \right] dt \\ &= \int_a^b \left[\left(P + R \frac{\partial z}{\partial x} \right) \frac{dx}{dt} + \left(Q + R \frac{\partial z}{\partial y} \right) \frac{dy}{dt} \right] dt \\ &= \int_{C_1} \left(P + R \frac{\partial z}{\partial x} \right) dx + \left(Q + R \frac{\partial z}{\partial y} \right) dy \\ &= \iint_D \left[\frac{\partial}{\partial x} \left(Q + R \frac{\partial z}{\partial y} \right) - \frac{\partial}{\partial y} \left(P + R \frac{\partial z}{\partial x} \right) \right] dA \end{aligned}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_D \left[\left(\frac{\partial Q}{\partial x} + \frac{\partial Q}{\partial z} \frac{\partial z}{\partial x} + \frac{\partial R}{\partial x} \frac{\partial z}{\partial y} + \frac{\partial R}{\partial z} \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} + R \frac{\partial^2 z}{\partial x \partial y} \right) \right. \\ &\quad \left. - \left(\frac{\partial P}{\partial y} + \frac{\partial P}{\partial z} \frac{\partial z}{\partial y} + \frac{\partial R}{\partial y} \frac{\partial z}{\partial x} + \frac{\partial R}{\partial z} \frac{\partial z}{\partial y} \frac{\partial z}{\partial x} + R \frac{\partial^2 z}{\partial y \partial x} \right) \right] dA \end{aligned}$$

EXAMPLE 1 Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = -y^2 \mathbf{i} + x \mathbf{j} + z^2 \mathbf{k}$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (Orient C to be counterclockwise when viewed from above.)

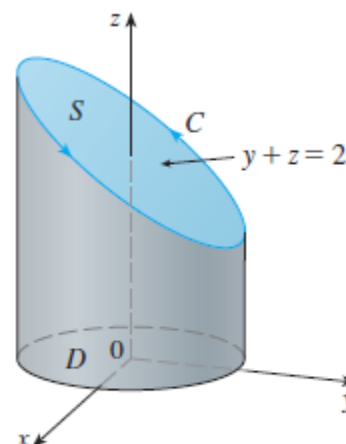


FIGURE 3

EXAMPLE 2 Use Stokes' Theorem to compute the integral $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xz \mathbf{i} + yz \mathbf{j} + xy \mathbf{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane. (See Figure 4.)

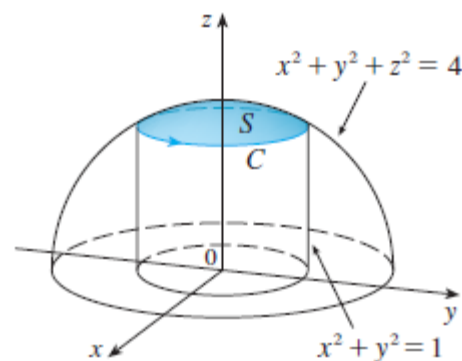
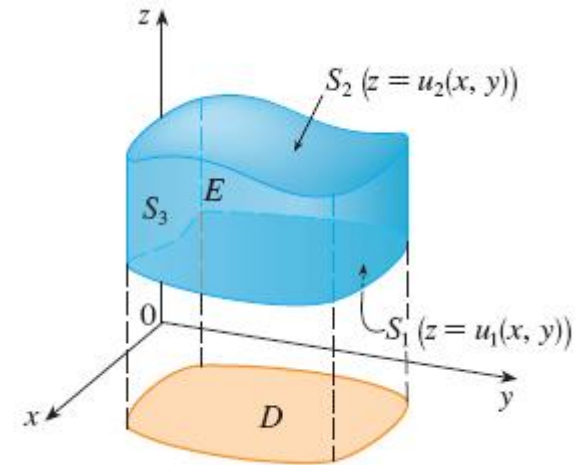


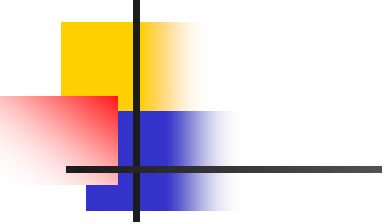
FIGURE 4

The Divergence Theorem



The Divergence Theorem Let E be a simple solid region and let S be the boundary surface of E , given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV$$



PROOF Let $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$. Then

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

so

$$\iiint_E \operatorname{div} \mathbf{F} \, dV = \iiint_E \frac{\partial P}{\partial x} \, dV + \iiint_E \frac{\partial Q}{\partial y} \, dV + \iiint_E \frac{\partial R}{\partial z} \, dV$$

If $\mathbf{n}$ is the unit outward normal of S , then the surface integral on the left side of the Divergence Theorem is

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S (P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}) \cdot \mathbf{n} \, dS \\ &= \iint_S P \mathbf{i} \cdot \mathbf{n} \, dS + \iint_S Q \mathbf{j} \cdot \mathbf{n} \, dS + \iint_S R \mathbf{k} \cdot \mathbf{n} \, dS \end{aligned}$$

$$\boxed{2} \quad \iint_S P \mathbf{i} \cdot \mathbf{n} \, dS = \iiint_E \frac{\partial P}{\partial x} \, dV$$

$$\boxed{3} \quad \iint_S Q \mathbf{j} \cdot \mathbf{n} \, dS = \iiint_E \frac{\partial Q}{\partial y} \, dV$$

$$\boxed{4} \quad \iint_S R \mathbf{k} \cdot \mathbf{n} \, dS = \iiint_E \frac{\partial R}{\partial z} \, dV$$

To prove Equation 4 we use the fact that E is a type 1 region:

$$E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$$

where D is the projection of E onto the xy -plane. By Equation 15.6.6, we have

$$\iiint_E \frac{\partial R}{\partial z} dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} \frac{\partial R}{\partial z}(x, y, z) dz \right] dA$$

and therefore, by the Fundamental Theorem of Calculus,

$$\boxed{5} \quad \iiint_E \frac{\partial R}{\partial z} dV = \iint_D [R(x, y, u_2(x, y)) - R(x, y, u_1(x, y))] dA$$

$$\iint_{S_3} R \mathbf{k} \cdot \mathbf{n} dS = \iint_{S_3} 0 dS = 0$$

The equation of S_2 is $z = u_2(x, y)$, $(x, y) \in D$, and the outward normal $\mathbf{n}$ points upward, so from Equation 16.7.10 (with $\mathbf{F}$ replaced by $R \mathbf{k}$) we have

$$\iint_{S_2} R \mathbf{k} \cdot \mathbf{n} dS = \iint_D R(x, y, u_2(x, y)) dA$$

On S_1 we have $z = u_1(x, y)$, but here the outward normal $\mathbf{n}$ points downward, so we multiply by -1 :

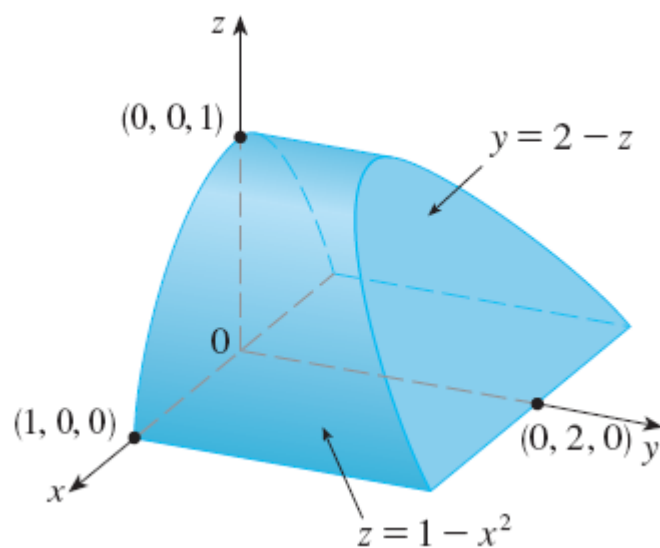
$$\iint_{S_1} R \mathbf{k} \cdot \mathbf{n} dS = - \iint_D R(x, y, u_1(x, y)) dA$$

V EXAMPLE 1 Find the flux of the vector field $\mathbf{F}(x, y, z) = z\mathbf{i} + y\mathbf{j} + x\mathbf{k}$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

V EXAMPLE 2 Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where

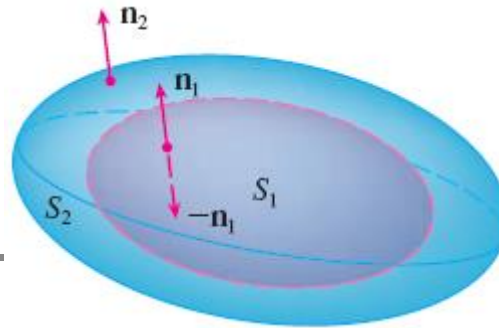
$$\mathbf{F}(x, y, z) = xy\mathbf{i} + (y^2 + e^{xz^2})\mathbf{j} + \sin(xy)\mathbf{k}$$

and S is the surface of the region E bounded by the parabolic cylinder $z = 1 - x^2$ and the planes $z = 0$, $y = 0$, and $y + z = 2$. (See Figure 2.)



$$\frac{184}{35}$$

Divergence theorem for regions enclosed by surfaces



Although we have proved the Divergence Theorem only for simple solid regions, it can be proved for regions that are finite unions of simple solid regions. (The procedure is similar to the one we used in Section 16.4 to extend Green's Theorem.)

For example, let's consider the region E that lies between the closed surfaces S_1 and S_2 , where S_1 lies inside S_2 . Let $\mathbf{n}_1$ and $\mathbf{n}_2$ be outward normals of S_1 and S_2 . Then the boundary surface of E is $S = S_1 \cup S_2$ and its normal $\mathbf{n}$ is given by $\mathbf{n} = -\mathbf{n}_1$ on S_1 and $\mathbf{n} = \mathbf{n}_2$ on S_2 . (See Figure 3.) Applying the Divergence Theorem to S , we get

7

$$\begin{aligned}\iiint_E \operatorname{div} \mathbf{F} \, dV &= \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS \\ &= \iint_{S_1} \mathbf{F} \cdot (-\mathbf{n}_1) \, dS + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 \, dS \\ &= -\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S}\end{aligned}$$